

# Solution Requirements

|              |                                          |               |                          |
|--------------|------------------------------------------|---------------|--------------------------|
| Date         | 30 September 2026                        | Team ID       | 6ab3e57da238dc999eb3d44d |
| Project Name | EduGenie – AI-Powered Learning Assistant | Maximum Marks | As per template          |

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**GitHub:** <https://github.com/ricardovictor-28507/Google-Cloud-Generative-AI--A-S>

## Step 1: Functional Requirements (FR)

| S.No | Requirement Category  | Requirement Description                                                     | Priority |
|------|-----------------------|-----------------------------------------------------------------------------|----------|
| 1    | Question answering    | Accept a learner question and return an AI-generated answer.                | High     |
| 2    | Explanation           | Generate a beginner-friendly explanation with an analogy and key takeaways. | High     |
| 3    | Summarization         | Convert supplied study text into a concise summary.                         | High     |
| 4    | Quiz generation       | Generate exactly three MCQs with four options and a matching answer field.  | High     |
| 5    | Learning path         | Generate a structured beginner-to-advanced study path.                      | High     |
| 6    | External AI interface | Connect to Google Gemini through the google-genai SDK.                      | High     |
| 7    | Validation            | Reject empty and oversized text/topic requests.                             | High     |
| 8    | Demo fallback         | Keep core UI/API testable without a live Gemini key.                        | Medium   |

## Step 2: Non-Functional Requirements

| Category        | Requirement                  | Acceptance Criteria                                                                                           |
|-----------------|------------------------------|---------------------------------------------------------------------------------------------------------------|
| Performance     | Responsive local interaction | No critical application errors during normal local use; benchmark values to be established before production. |
| Reliability     | Graceful AI/API failure      | Fallback or useful error is shown instead of silent failure.                                                  |
| Security        | Protect API credentials      | Key stored in <code>.env</code> ; <code>.env</code> excluded from version control.                            |
| Maintainability | Modular code                 | Each learning workflow has a focused module and shared AI service.                                            |
| Usability       | Simple learner workflow      | Five clearly labeled tools with readable results.                                                             |
| Scalability     | Allow future feature growth  | New modules/endpoints can be added without changing the shared AI configuration.                              |